CSE 4345 Big Data Analytics • CEP EpiScope • Group of four

Status and tasks

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Where we are

The pipeline runs end to end and passes 141 automated tests. The hidden-drop drill passed from a clean clone on 11 Oct at SF=0.05: a new 14-day drop added 8,292 rows, a re-run added no duplicates, and the dashboard data was 9.64 MB with no counts from 1 to 4. What is missing is mostly evidence and writing: the scaling study, the Spark optimisation measurements, the detection and backtest result files, the usability test, the report and the video. All 14 commits so far are Shezan's and fall on one day; R9 grades each person's own Git share, so **everyone must commit their own work from now on**.

The deadline has not been announced. Plan as if it is a few days away.

Every requirement

ID	Pri	Owner	Status	Evidence or gap
ING-1	M	Shezan	done	generator plus bronze with checksum, batch ID, load time; path bug fixed 20 Oct
ING-2	M	Shezan	done	every heavy transform in Spark
ING-3	M	Shezan	done	Delta by `epi_year`; Hive catalog step added 20 Oct (`make catalog`)
ING-4	M	Shezan	done	Delta MERGE keeps the latest report; tests and drill show no duplicates
ING-5	S	Shezan	missing	skipped on purpose (RAM); the report says why
ING-6	S	Shezan	missing	streaming path promised in the proposal, not built yet
ING-7	C	Shezan	missing	Delta history exists; "as of date t" reader not built yet
DQ-1	M	Shezan	done	14 encounter, 7 facility, 7 stock rules; quarantine with reason codes
DQ-2	M	Shezan	done	`gold.dq_report` and `dq.json` on the dashboard
DQ-3	M	Shezan	done	`batch_ids` on every gold row; lineage test
DQ-4	S	Shezan	done	precision 0.998 on 441 flagged pairs; recall 0.11 (stated honestly)
PRV-1	M	Samira	done	HMAC-SHA256 from Spark `sha2`; key only in `.env`
PRV-2	M	Rimjhim	done	complementary suppression with a leak audit in every export
PRV-3	S	Rimjhim	missing	no role-based (national or district) view

ID	Pri	Owner	Status	Evidence or gap
PRV-4	C	Samira	partial	k-anonymity check exists; differential-privacy text still to write
AN-1	M	Rafi	done	EARS C2 (window functions) and CUSUM (`applyInPandas`)
AN-2	M	Samira	partial	evaluation code and tests; no result file committed
AN-3	S	Rafi	done	onset-week counting and completeness; numbers wait for the evaluation run
AN-4	M	Rafi	done	per-district NB-GLM with lagged weather
AN-5	M	Samira	partial	backtest code; no result file committed
AN-6	M	Rafi	done	demand conversion with stated assumptions
AN-7	M	Rafi	partial	LP works, but the only run so far had almost no net need (see Rafi)
AN-8	S	Rafi	partial	curve over 5 budgets and 11 values of α; recommends $\alpha = 0$ everywhere
AN-9	M	Rafi	done	STL calendar per division and disease
AN-10	C	Rafi	missing	SEIR only inside the generator
VIS-1–7	M/S/C	Rimjhim	done	D3 v7 only, five linked views, what-if panel, mobile layout
Usability	M	Rimjhim	missing	test with at least two non-CS users
ENG-1	M	Samira	partial	`make setup && make all` works; README lacks prerequisites, drop steps, architecture
ENG-2	M	Samira	missing	scaling at SF 0.25, 0.5, 1; harness `scripts/scaling.sh` ready
ENG-3	M	Samira	missing	two optimisations with before and after Spark UI screenshots
ENG-4	S	Samira	partial	141 tests; no DAG with retries
Report		all	draft	skeleton in `report/` (02 in this pack)
Video		all	elsewhere	slides and script built separately; links coming

Shezan (Data Engineer)

Task	Effort	Files
Run the scaling harness at SF 0.25, 0.5 and 1 (feeds Samira's ENG-2 and gate 5)	2 h of runs	`scripts/scaling.sh`
ING-6 streaming path that refreshes alerts within 5 minutes	2 h	`pipeline/stream.py` (new)
ING-7 "what did we know on date t" reader on Delta history	45 min	`pipeline/bronze/lake.py`
README: prerequisites, hidden-drop steps, architecture summary	30 min	`README.md`
Final hidden-drop drill on the last commit	30 min	`scripts/hidden_drop_drill.sh`
Report §4 (data generation, quality, lineage), own reflection	2 h	`report/sections/`

Tahsin Ahmed Rafi (Analytics)

Task	Effort	Files
Check the allocation: in the SF=0.05 run the national net need was 0 for beds, saline, NS1 kits and ORS (stock exceeded demand), so the LP had nothing to allocate. Recheck on the final data; consider planning over the season peak or a lower starting stock	1–2 h	`analytics/optimise/`, `analytics/forecast/demand.py`
AN-8: the recommender picks $\alpha = 0$ at every budget; make sure the curve supports a real choice	1 h	`analytics/optimise/curve.py`
AN-3 figure: naive detector against ours on the artefact facility	1 h	`analytics/evaluation/`
AN-10 SEIR module (bonus, optional)	2 h	`analytics/scenario/` (new)
Report §5 and the analytics part of §6, own reflection	3 h	`report/sections/`

Rimjhim Dey (Visualisation)

Task	Effort	Files
Usability test with at least two non-CS users: tasks, what they did, what we changed	1 h plus fixes	`docs/usability.md` (new)
PRV-3 role-based view: national sees all; a district user sees their own district in detail	2 h	`dashboard/js/state.js`, `serving/export.py`
Dashboard screenshots for the report (all five views, what-if, suppressed cells)	30 min	`report/figures/`
Report §7, own reflection	2 h	`report/sections/`

Samira (Platform & Evaluation, report lead)

Task	Effort	Files
ENG-2 write-up: runtime per stage, sizes, bottleneck from the Spark UI (runs from Shezan's harness)	1 h	`docs/experiments/scaling.json`
ENG-3: two optimisations with before and after numbers and Spark UI screenshots (for example broadcast of the register, partition pruning on `epi_year`, AQE skew on Dhaka)	2 h	`docs/experiments/eng3/` (new)
ENG-4: DAG with retries (the proposal promised it)	1 h	`orchestration/`
AN-2 and AN-5: run `make evaluate` and the backtest on the final data and commit the result files	1 h plus run time	`analytics/evaluation/`, `analytics/forecast/backtest.py`
Report lead: §1, §2, §3, §6 (scaling), §8, §9, policy brief (Appendix A), contribution matrix	5 h	`report/`

Rules for everyone

- Commit your own work under your own Git identity, in small commits that start with the requirement ID, for example `AN-8: ...`. Nobody commits in someone else's name.
- Never commit `.env`, a salt or a key. Data stays out of Git (`data/` and `lake/` are ignored).
- Every number in the report and the video comes from a file in `docs/experiments/` of the final run.
- Nobody tags `v1.0` or records the video until the group agrees the code is frozen.
- Write your own half-page reflection (Appendix C) and agree the contribution matrix (Appendix B) together.